CCNA – DAY 6
STATIC ROUTING

Learning Notes • Concepts • Cisco CLI • Practice • Interview Questions

1. What is Routing?

Routing is the process of selecting a path to send packets from one network to another. A router checks its routing table and forwards packets towards the destination network. In simple words, routing helps different networks communicate.

2. What is Static Routing?

Static routing is a method in which an administrator manually configures a route on a router. The route does not automatically change when the network topology changes; the administrator must update it.

Example: R1 needs to reach network 192.168.2.0/24 through the next-hop router 10.0.0.2.

3. Static Route Syntax

Cisco IOS command:

```
ip route <destination-network> <subnet-mask> <next-hop-ip | exit-interface>
```

Example:

```
ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Meaning: To reach 192.168.2.0/24, forward packets to next-hop IP 10.0.0.2.

4. Simple Network Example

Device / Network	IP address / Network
R1 LAN	192.168.1.0/24
R1–R2 link	10.0.0.1/30 (R1) and 10.0.0.2/30 (R2)
R2 LAN	192.168.2.0/24

R1 configuration – route to R2 LAN:

```
R1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

R2 configuration – route to R1 LAN:

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Important: Configure a return route too. A route in only one direction may not allow two-way communication.

5. Default Static Route

A default route is used when the router has no more specific route for a destination. It is often used to send unknown traffic towards an ISP or upstream router.

```
ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

0.0.0.0/0 means any IPv4 destination not matched by a more specific route.

6. Verify and Troubleshoot

Command	Purpose
show ip route	Displays the IPv4 routing table.
show ip interface brief	Checks interface status and IP addresses.

ping 192.168.2.1	Tests reachability to a destination.
traceroute 192.168.2.1	Shows the Layer 3 path towards a destination.
show running-config | include ip route	Displays configured static-route lines.

Troubleshooting checklist:

• Check that router interfaces are up/up and have correct IP addresses.

• Check the destination network and subnet mask in the static route.

• Verify that the next-hop IP is reachable.

• Check that the remote router has a return route.

• Use ping and traceroute to identify where connectivity fails.

7. Static Routing vs Dynamic Routing

Static Routing	Dynamic Routing
Routes are manually configured.	Routers learn routes using routing protocols.
Simple for small, stable networks.	Useful for larger or changing networks.
No routing-protocol updates are exchanged.	Routers exchange routing information.
Administrator maintains route changes.	Routes can adjust automatically based on protocol rules.

8. Interview Questions and Answers

Q1. What is static routing?

A routing method where an administrator manually adds routes to a router.

Q2. Which command configures a static route?

ip route destination-network subnet-mask next-hop-ip (or exit interface).

Q3. What is a default route?

A route used when no more specific route exists. For IPv4, it is 0.0.0.0/0.

Q4. What does S mean in show ip route?

It identifies a static route in the IPv4 routing table.

Q5. Main disadvantage of static routing?

It requires manual maintenance and does not automatically adapt to topology changes.

Q6. Why is a return route needed?

The destination network also needs a path back for two-way communication.

Q7. How do you check the routing table?

Use show ip route.

9. Practice Questions

1. Configure a static route on R1 to reach 172.16.2.0/24 through next hop 10.10.10.2.

2. Write the IPv4 default-route command using next hop 192.168.1.1.

3. Which command displays the routing table?

4. Two LANs cannot communicate. List four things you would check.

5. Explain why both routers may need static routes in a two-router lab.

Answer Key

1. ip route 172.16.2.0 255.255.255.0 10.10.10.2
2. ip route 0.0.0.0 0.0.0.0 192.168.1.1
3. show ip route
4. Interface status/IP, route network and mask, next-hop reachability, and return route.
5. Each router needs a path to the remote network so traffic can travel in both directions.

Day 6 learning goal: Understand static routes, configure a default route, verify the routing table, and troubleshoot basic connectivity in Packet Tracer.

Study tip: First understand the network diagram, then configure both routers, and finally test with ping.